yx49k-L20: A 308M Hybrid Gated-Delta-Attention Language Model Trained on 367B Tokens on TPU v4

David-Franco Buga (Yxanul)
github.com/yxanul · huggingface.co/Yxanul

August 2026

Abstract

We describe the complete construction of a 307.7M-parameter hybrid language model trained end-to-end on TPU Research Cloud hardware: a gated-delta-attention (KDA) architecture with per-channel decay interleaved 3:1 with NoPE global attention; custom fused Pallas training kernels built around a chunkwise WY representation, a divide-and-conquer nilpotent inverse, and an overflow-bounded decay factorization that turns the gate's lower bound into a kernel compute parameter; a precision policy qualified against fp64 on real text; a matrix/scalar split optimizer with orthogonalized updates and a logit-growth guard; and a 49,152-token tokenizer constructed as an exact subset projection of a 4B-parameter teacher's vocabulary, which later makes logit distillation alignment-free by construction. The model was pretrained on 367B tokens of ClimbMix on a 32-chip TPU v4-64 at a median 1.09M tokens/s, surviving a mid-anneal crash with a verified clean resume. At 0-shot it reaches HellaSwag 56.9, PIQA 75.2, ARC-e 67.5, and WinoGrande 59.4 — ahead of a same-size reference model trained on 11× more tokens on every comparable benchmark. A post-training stage of supervised fine-tuning plus offline top-K logit distillation lifts instruction following to IFEval 51.0, above both major same-size instruction-tuned references when all models are measured under an identical harness. We close with a measured attribution of what remains: capability favors our base; decode-time discipline favors the references' deeper post-training; and the failure signature identifies on-policy distillation as the indicated next stage.

Contents

1 Introduction

This report documents a complete small-language-model program carried out on TPU Research Cloud hardware between 2026-07-20 and 2026-08-08: kernel development on a single-host TPU v6e-8, a 367B-token pretraining campaign on a 32-chip TPU v4-64, and a post-training stage (supervised fine-tuning plus offline logit distillation) evaluated against the strongest publicly available models of the same size class, all measured under one harness.

The program's through-line is that every layer of the stack was co-designed with the ones above and below it. The architecture's gate bound is also the kernel's floating-point overflow budget (§3). The tokenizer is not merely small; it is an exact subset projection of a 4B teacher's vocabulary, which later deletes an entire class of distillation alignment machinery (§6, §9). The optimizer's logit-growth guard is priced at zero because its trigger never fired in 700,000 steps — a fact we can state only because the training loop recorded per-head maximum logits the whole way (§5).

Contributions.

1. A hybrid gated-delta-attention (KDA) model, `kda_hybrid_yx49k_l20`: 307,698,680 parameters, 20 layers as five $[\mathrm{KDA}, \mathrm{KDA}, \mathrm{KDA}, \mathrm{GQA}]$ cycles, trained at sequence length 2048 (§2).

2. Fused TPU training kernels for KDA built on a chunkwise WY form: a divide-and-conquer explicit inverse that is exact by nilpotency, a decay-anchored dot-product factorization with a provable fp32 overflow bound, and a split-backward variant for the older v4 ISA (§3), with a numerics policy qualified against fp64 on real text (§4).

3. A matrix/scalar split optimizer — orthogonalized momentum for matrices, AdamW for the rest — with a grouped-query adaptation of attention-logit clipping (§5).

4. The yx49k tokenizer: 49,152 tokens, each mapping to a distinct token of the teacher vocabulary, at practical parity with a 2.6× larger reference tokenizer on bytes-per-token (§6).

5. The campaign itself: 700k steps at 524,288 tokens per step on 32 chips, median 1.09M tokens/s, a mid-anneal crash with a measurably clean resume, and a terminal anneal worth −0.195 nats of holdout loss (§7, §8).

6. A post-training arc — SFT at 6.27M rows, then offline top-K logit distillation from a frozen 4B teacher — with every design decision A/B-measured, including a clean null result on repeated epochs (§9), and an external calibration against Gemma 3 270M IT and SmolLM 360M v1/v2 under identical measurement (§10).

Table 1: Summary card.

Parameters	307,698,680 (tied embeddings)
Layers	$20 = 5 \times$ [KDA, KDA, KDA, GQA], $d_{\text{model}} = 1024$
Tokenizer	yx49k (49,152 BPE, exact teacher subset)
Pretraining data	ClimbMix, 367B step-tokens (∼341B unique)
Hardware	TPU v4-64: 32 chips / 8 hosts, pure data parallel
Throughput	median 1.09M tokens/s (480 ms / 524,288-token step)
Final loss	train 2.3617, holdout 2.3494 (ppl 10.48)
0-shot	HellaSwag 56.9, PIQA 75.2, ARC-e 67.5, WinoGrande 59.4
Post-training	IFEval mean 51.0 (GOLD mix300k checkpoint)

2 Model architecture

2.1 Gated delta attention with per-channel decay

The sequence mixer used in three of every four layers is a linear attention layer whose state update follows the *delta rule*: rather than accumulating outer products indiscriminately, each write first subtracts what the state already predicts for the incoming key. With a scalar forgetting factor this is

$$S_t \;=\; \alpha_t \left(I - \beta_t k_t k_t^\top\right) S_{t-1} \;+\; \beta_t k_t v_t^\top, \qquad o_t = S_t^\top q_t, \tag{1}$$

where $S_t \in \mathbb{R}^{d_k \times d_v}$ is a per-head fast-weight matrix, $\beta_t \in (0,1)$ a learned write strength, and $\alpha_t \in (0,1]$ a learned decay. KDA generalizes the decay from a scalar to a *per-channel* vector $\alpha_t \in (0,1]^{d_k}$:

$$\boxed{S_t \;=\; \left(I - \beta_t k_t k_t^\top\right) \mathrm{Diag}(\alpha_t)\, S_{t-1} \;+\; \beta_t k_t v_t^\top, \qquad o_t = S_t^\top q_t.} \tag{2}$$

Each key channel now has its own forgetting timescale, so one head can simultaneously hold long-lived and rapidly-refreshed features. This is the modeling win that motivates the entire kernel effort of §3: per-channel decay breaks the algebraic shortcuts available to scalar-decay linear attention, and the training-time chunkwise form must track a full decay *profile* per position.

2.2 The bounded gate

The decay is produced in log space by a low-rank head with a hard lower bound. With $A = \exp(A_{\log}) \in \mathbb{R}^H$ a per-head positive rate and $f(x) = W^{\uparrow}(W^{\downarrow}x) + b_{dt}$ a rank-128 projection of the token's hidden state,

$$\log \alpha_t \;=\; g_t \;=\; \ell \cdot \sigma\big(A \odot f(x_t)\big), \qquad \ell = -5, \tag{3}$$

so that $g_t \in (-5, 0)$ always. An unbounded softplus form $g_t = -A\,\mathrm{softplus}(f(x_t))$ is implemented as an alternative, but the bounded form is required by the production kernel: the bound $|g| < 5$ is exactly what makes the kernel's decay-factorized dot products representable in fp32 (§3.2). The write strength is a per-head scalar $\beta_t = \sigma(w_\beta^\top x_t)$.

2.3 The full layer

Around the recurrence, the layer applies a fused QKV projection, a short causal depthwise convolution (width w=4, one filter per channel) with a SiLU activation,

$$\tilde{x}_t = \mathrm{silu}\Big(\sum_{j=0}^{w-1} W_j^{\mathrm{conv}} \odot x_{t-(w-1)+j}\Big), \qquad (q_t, k_t, v_t) = \mathrm{split}(\tilde{x}_t), \tag{4}$$

then ℓ_2-normalizes q and k per head ($\hat{u} = u/\max(\|u\|_2, \varepsilon)$, with $q \leftarrow \hat{q}/\sqrt{d_k}$), runs (2), and finishes with a per-head RMSNorm modulated by a learned low-rank output gate:

$$y_t = W_o\left(\mathrm{RMSNorm}(o_t) \odot \sigma\big(W_g^{\uparrow}(W_g^{\downarrow}x_t) + b_g\big)\right). \tag{5}$$

Heads are $H = 8$ with $d_k = d_v = 128$.

2.4 The hybrid stack

Layers are arranged in cycles of four — three KDA layers followed by one full-attention layer — repeated five times for the 20-layer model. The global layer is grouped-query attention (8 query heads, 2 KV heads, $d_h = 128$) with *no positional encoding* (NoPE): position information reaches it only through the recurrent layers below and the causal convolutions, which measurably suffices at this scale and keeps the decode-time KV cache free of any position-dependent transform. The channel mixer everywhere is a fused SwiGLU MLP, $\mathrm{MLP}(x) = W_o\big(\mathrm{silu}(W_1x) \odot W_2x\big)$ with inner width 2816; normalization is pre-norm RMSNorm ($\varepsilon = 10^{-5}$, no centering, no bias); the output head is tied to the input embedding. Cycles are scanned with parameters stacked along a scan axis, and rematerialization operates per cycle.

Table 2: `kda_hybrid_yx49k_l20` hyperparameters.

vocabulary	49,152	layers	20 (5 cycles)
d_{model}	1024	cycle	[KDA, KDA, KDA, GQA]
MLP inner	2816 (SwiGLU)	norm	RMSNorm, ε=10^{-5}
KDA heads	8 × (d_k=d_v=128)	conv width	4
gate	rank 128, $\ell = -5$	q, k	ℓ_2-normalized
GQA heads	8 query / 2 KV, d_h=128	positions	NoPE
embeddings	tied	sequence	2048
precision	bf16 compute, fp32 master	params	307,698,680

Parameter accounting: the tied embedding contributes $49{,}152 \times 1024 = 50.3$M; each cycle carries three KDA mixers, one GQA mixer, four SwiGLU blocks, and their norms; five cycles plus the final norm bring the total to 307,698,680. Relative to same-size references, the small vocabulary shifts roughly 25M parameters from the embedding table into the transformer body — a deliberate exchange of rare-token coverage for depth-adjusted compute, made survivable by the tokenizer's byte-level fallback (§6).

3 Chunkwise WY form and the TPU kernels

Equation (2) is a sequential recurrence; trained naively it serializes 2048 steps. The training path instead processes the sequence in chunks of $C = 64$, expressing everything inside a chunk with dense matrix algebra (which TPUs execute at full matrix-unit utilization) and carrying the fast-weight state only across chunk boundaries.

3.1 Chunkwise decomposition

Write $G_i = \sum_{s \le i} g_s \in \mathbb{R}^{d_k}$ for the inclusive cumulative log-decay within a chunk. Unrolling (2) from the chunk-entry state S gives, for the output at intra-chunk position i, a *cross-chunk* term read through the accumulated decay and an *intra-chunk* term over earlier positions of the same chunk:

$$o_i \;=\; \big(q_i \odot e^{G_i}\big)^{\top} S \;+\; \sum_{j\le i} \Xi_{ij}\,\hat{v}_j, \qquad \Xi_{ij} = \big\langle q_i \odot e^{G_i - G_j},\, k_j \big\rangle, \tag{6}$$

where the $\hat{v}_j$ are *delta-corrected* values: the values actually written to the state after subtracting the state's own prediction. The corrections inside one chunk satisfy a unit-lower- triangular linear system. Defining the decay-weighted Gram matrix $M_{ij} = \langle \beta_i k_i \odot e^{G_i - G_j},\, k_j\rangle$ for $j < i$ and $T = \mathrm{StrictLower}(M)$,

$$(I + T)\,U = B_v, \qquad (I + T)\,W = B_k, \tag{7}$$

with $B_v = [\beta_i v_i]_i$ and $B_k = [\beta_i k_i \odot e^{G_i}]_i$. This is the WY representation of the product of rank-one delta updates: U and W compress the chunk's 64 sequential writes into two dense matrices. The corrected values and the state advance are then

$$\hat{V} = U - WS, \qquad S \leftarrow \mathrm{Diag}\big(e^{G_C}\big)\, S + \big(k \odot e^{G_C - G}\big)^{\top} \hat{V}, \tag{8}$$

one dense contraction per chunk, with S the only sequential carrier.

3.2 Decay-anchored pairwise dots and the overflow bound

Every matrix above contains entries of the form

$$M_{ij} \;=\; \sum_d L_{i,d}\, R_{j,d}\, e^{G_{i,d} - G_{j,d}}, \tag{9}$$

which cannot be computed as a plain matmul because the decay factor depends on *both* indices and on the channel. Materializing the $[C, C, d_k]$ tensor is prohibitive; factorizing $e^{G_i - G_j} = e^{G_i} e^{-G_j}$ overflows, since $-G_j$ can reach $|\ell| \cdot C = 320$ and e^{320} is far beyond fp32.

The kernel factorizes against a moving *anchor* instead. Rows are processed in blocks of R rows; each block is anchored at its last row's cumulative decay $a = G_{\text{block end}}$:

$$M_{ij} = \sum_d \underbrace{\big(L_{i,d}\, e^{G_{i,d} - a_d}\big)}_{\text{left factor}} \cdot \underbrace{\big(R_{j,d}\, e^{a_d - G_{j,d}}\big)}_{\text{right factor}}, \tag{10}$$

a true matmul per block. The anchor cancels exactly in the product. The right factor's exponent $a_d - G_{j,d}$ is nonpositive for every valid column (j at or before the anchor), so the right factor never exceeds 1; the left factor's exponent spans at most the R rows of one block. Since $g \in (\ell, 0)$ with $\ell = -5$ (Eq. 3), the one-sided exponent obeys

$$\max_{i,d} |G_{i,d} - a_d| \;\le\; |\ell| \cdot R \;=\; 5 \cdot 16 = 80, \qquad e^{80} \approx 5.5 \times 10^{34} \;<\; 3.4 \times 10^{38} = \text{fp32 max}. \tag{11}$$

This inequality is checked at kernel import time. It is why the gate bound $\ell = -5$ is a *kernel compute parameter*: relaxing the gate below -5, or widening the row block past 16 with last-row anchoring, breaks (11) and the kernel refuses to build. (Anchoring at the block midpoint would halve the one-sided exponent and double the admissible block; the production kernels use $R = 8$ on v5e/v6e and $R = 16$ on v4, both comfortably inside the bound.) Invalid future columns are set to $-\infty$ *before* exponentiation — masking after the fact would leave overflowed values in the backward graph.

3.3 The WY solve: a divide-and-conquer nilpotent inverse

System (7) is a unit-lower-triangular solve of size $C = 64$ against a 256-wide right-hand side. Forward substitution is stable but latency-bound on a TPU: 60 chained small matmuls per solve. The production kernel forms the explicit inverse instead, by divide and conquer. Split

$$I + T = \begin{pmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{pmatrix}, \qquad (I + T)^{-1} = \begin{pmatrix} A_{11}^{-1} & 0 \\ -A_{22}^{-1} A_{21} A_{11}^{-1} & A_{22}^{-1} \end{pmatrix}. \tag{12}$$

Writing M for the block-diagonal matrix of the two inverted halves and $C_\times$ for the pre-multiplied coupling block, the merge step is the closed form

$$\boxed{(I+T)^{-1} \;=\; M - M\,C_\times\,M} \tag{13}$$

Proposition 1. *Equation* (13) *is exact, not a truncation.*

Proof. $MC_\times M$ has only a lower-left block, so $(MC_\times M)C_\times(M\cdot) = 0$: the coupling is nilpotent of index two, and the Neumann series $M - MC_\times M + MC_\times MC_\times M - \cdots$ terminates after its second term. □

Base blocks of size 2 invert as $I - T_2$ with no arithmetic at all; five merge levels build the full 64×64 inverse in ten uniform matmuls, and applying it to the right-hand side is a single dense matrix-unit contraction. Two alternatives were implemented and rejected by measurement: blocked forward *substitution* (stable, −30% end-to-end throughput) and a *repeated-squaring* Neumann form whose intermediate powers $T^2, T^4, \ldots, T^{32}$ grow to $\sim 10^{12}$ on real-text keys and destroy the backward pass by cancellation (§4).

3.4 The fused kernels

The production kernel fuses the entire mixer — causal conv, SiLU, normalization, cumulative decay, pairwise dots, the solve, and the chunk scan — into one Pallas program per (batch, chunk-stream): grid $(B, T/C)$ with the chunk axis sequential, all heads processed together, the $H \times 128 \times 128$ fp32 state and the $w-1$ rows of convolution history resident in VMEM across chunk iterations. XLA never materializes a convolved, activated, split, or head-transposed copy of the QKV projections; the kernel consumes the raw $[B, T, 3, H, D]$ projection output and emits bf16 outputs plus an fp32 state history (one snapshot per chunk) for the backward.

The backward is a second fused kernel running the grid in reverse: it recomputes each chunk's forward quantities from the raw projections, carries $\bar{S}$ in VMEM, threads the $w-1$ *future* rows of convolution cotangents backward through a scratch ring, and applies hand-derived adjoints for every stage — including a blockwise VJP of (9) that never materializes the $[C, C, d_k]$ tensor, the solve adjoint $\bar{T} = -\mathrm{tril}\big(A^{-\top}\bar{A}A^{-\top}, -1\big)$, and a reverse cumulative sum taking $\bar{G}$ back to $\bar{g}$.

TPU v4 variant. The folded kernel's in-kernel $[\mathrm{time}, \mathrm{head}]$ transposes lower to sublane gathers that the v4 compiler backend rejects, so v4 runs a pre-fold variant: the QKV mixer stays in XLA and the backward is *split into two kernels* at the solve/pairwise boundary — a reverse sequential stage exporting per-chunk partials to HBM, then a chunk-parallel epilogue. On the campaign hardware the three KDA kernel launches together account for ∼15% of the training step; the dense GEMM body dominates.

Table 3: Kernel generations on the 273M-class hybrid, TPU v6e-8, batch 8 × 2048. "Core" is the isolated KDA forward+backward.

	core fwd+bwd	full-model throughput
analytical XLA backward	31.9 ms	514k tok/s (core)
fused Pallas, substitution solve	—	473k tok/s
fused Pallas, divide-and-conquer inverse	5.0 ms	**616k tok/s**
blanket one-pass bf16 (rejected)	4.9× faster core	NaN by step 2

4 Numerics and the precision policy

TPU matrix units evaluate an "fp32" matmul as one, three, or six decomposed bf16 passes. The kernel's policy, called `guarded_fp32`, spends the passes exactly where the algebra is fragile and nowhere else:

Two measurements fix the policy's shape. First, running *every* stage at one pass is 4.9× faster in the isolated core and destroys the model: the loss reaches NaN by the second optimizer step. Guarding

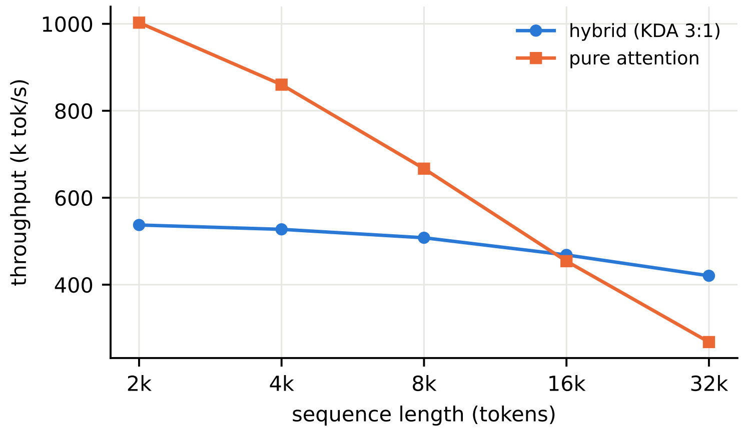

Figure 1: Throughput vs. sequence length, hybrid (3:1 KDA) against a pure-attention control of the same size (TPU v6e-8). The recurrent layers' linear scaling overtakes quadratic attention near 16k tokens; at 32k the hybrid is 1.6× faster. At the campaign's 2048 the hybrid pays ∼2× for its richer mixer — the price of buying long-context headroom at training time.

Table 4: Per-stage precision inside the fused kernels.

stage	storage	matmul passes
q, k, v streams	bf16	1
fast-weight state S, cumulative decay G	fp32	1
pairwise dots (9), chunk scan	fp32 accum	1
WY inverse formation (13)	fp32	6 (v6e) / 3 emulated (v4)
inverse application, solve adjoints	fp32	6 (v6e) / 3 emulated (v4)
logits, softmax, loss	fp32	—
master weights, optimizer state	fp32	—

only the solve recovers a stable loss curve at 561k tok/s — 91% of the full fused speedup. Second, the rejected repeated-squaring solve (§3.3) is a cautionary tale about validating on real data: on random keys its intermediate powers stay tame, but correlated real-text keys drive $\|T^{2^j}\|$ toward 10^{12}, and although the *forward* answer survives, the backward pass loses to catastrophic cancellation — gradient norm 3,933.7 against a reference of 2.407. The divide-and-conquer inverse at three emulated passes closes to a gradient norm of 2.406743 vs. the fp32 reference 2.406825 (relative ℓ_2 error 1.86% at cosine 0.99983), and a per-layer sweep of solve condition numbers on checkpoint activations ($\kappa_2 \leq 10.4$, solve error $\leq 2.3 \times 10^{-5}$ against fp64) qualified the 3-pass setting for the campaign.

The v4 3-pass mode is emulated in-kernel: each fp32 operand splits into a bf16 high part and a bf16 residual, the three significant cross products accumulate in fp32, and the dropped low×low term is bounded near 2^{-16} relative — with a one-line rollback to six passes kept in the source. Stability is a necessary but not sufficient claim: the policy's residual 1.86% gradient error is *monitored*, not assumed away, and the campaign's loss/gradient telemetry (§7) is the ongoing evidence that it never mattered.

5 Optimizer: orthogonalized updates with a logit guard

The optimizer, called *MuonClip* in the codebase, routes every parameter by role: two-dimensional "matrix-like" parameters (the KDA projections, GQA QKV and output, both SwiGLU matrices) receive orthogonalized momentum updates; everything else (embeddings, the tied logit head, norm scales, biases, depthwise conv filters, the gate's per-head scalars) receives AdamW. Routing is fail-closed: a parameter with no declared role, or a matrix parameter without declared factorization axes, is a build error rather than a silent default.

5.1 The matrix update

For a matrix gradient M (after global clipping and momentum $\beta = 0.95$ with Nesterov), the update direction is the approximate orthogonalization of M by a Newton–Schulz iteration on the normalized matrix $X_0 = M/(\|M\|_F + \varepsilon)$:

$$X_{n+1} \;=\; aX_n + \big(b\,A_n + c\,A_n^2\big)X_n, \qquad A_n = X_n X_n^\top, \qquad (a,b,c) = (3.4445,\, -4.7750,\, 2.0315), \tag{14}$$

run for five iterations. The polynomial's fixed coefficients push every singular value of X toward 1, so the update has near-uniform spectral footprint — the property that lets small models take unusually large stable learning rates. The applied update is scaled for consistent root-mean-square magnitude across shapes,

$$\Delta W \;=\; -\,\eta \cdot 0.2\sqrt{\max(\mathrm{fan}_{\mathrm{in}}, \mathrm{fan}_{\mathrm{out}})} \cdot \mathrm{NS}_5(M), \tag{15}$$

which corrects a 6–10× effective learning-rate undershoot that the default width-scaling rule produces at these shapes. Scanned cycles treat the scan axis as a batch of independent orthogonalization problems; decoupled weight decay ($\lambda = 0.1$) and the learning-rate schedule are shared with the AdamW arm ($\beta_1 = 0.9$, $\beta_2 = 0.95$, nesterov), and global gradient clipping to norm 1 runs before either arm in fp32.

5.2 QK-Clip for grouped-query attention

Orthogonalized updates are known to permit slow unbounded growth of attention logits over very long runs. As insurance, after each update the fused QKV kernel of every cycle's attention layer is rescaled whenever any head's maximum pre-softmax logit $s^{\max}_{c,h}$ (recorded on-device across the global batch) exceeds a threshold $\tau = 100$:

$$\gamma_{c,h} = \min\left(1,\ \frac{\tau}{s^{\max}_{c,h} + \epsilon}\right), \qquad W_q^{(h)} \mathrel{*}= \sqrt{\gamma_{c,h}}, \qquad W_k^{(j)} \mathrel{*}= \sqrt{\min_{h \in \mathrm{group}(j)} \gamma_{c,h}}. \tag{16}$$

The grouped-query structure requires the adaptation on the right: a KV head shared by several query heads takes the group's most conservative scale, so the product of applied factors never exceeds any member head's γ. Values and optimizer state are untouched.

The guard's cost–benefit is now measurable: across all 700,000 campaign steps the recorded maximum attention logit peaked at **27.5** — a factor 3.6 below the threshold. The clip never fired. It cost one reduction and one conditional rescale per step and bought the certainty that the run could not be lost to logit blow-up; insurance that never paid out, priced at approximately zero.

5.3 Learning-rate schedule

The campaign schedule is warmup–constant–anneal:

$$\eta(t) = \begin{cases} \eta_{\max}\, t/t_w & t < t_w \quad \text{(brief linear warmup)} \\ \eta_{\max} & t_w \le t < T - T_a \\ \eta_{\max} \cdot \frac{1}{2}\big(1 + \cos \pi \frac{t-(T-T_a)}{T_a}\big) & t \ge T - T_a, \end{cases} \tag{17}$$

with $\eta_{\max} = 2 \times 10^{-3}$, total steps $T = 700\text{k}$ and a terminal anneal of $T_a = 70\text{k}$ steps. The constant plateau keeps the option of extending the run open until the end; the entire convergence benefit of decay is then collected in the final 10% of the budget (§7). The peak value itself came out of an earlier sweep on the predecessor campaign, which found the then-current setting roughly 10× under-tuned — with consistent-RMS scaling (15), 2×10^{-3} is comfortably stable for a 308M model.

6 The yx49k tokenizer: an exact subset of the teacher

Most small models inherit a large general-purpose tokenizer and spend a third or more of their parameters on the embedding table. We built a 49,152-token BPE with one additional structural requirement chosen for what comes after pretraining: *every* student token must correspond to a distinct token of the Qwen3.5 vocabulary (the tokenizer of the 4B model later used as a distillation teacher).

Formally, with student vocabulary V_s ($|V_s| = 49{,}152$) and teacher vocabulary V_t ($|V_t| = 248{,}077$), the construction fixes an injective map

$$m : V_s \hookrightarrow V_t \qquad \text{such that } \mathrm{bytes}(i) = \mathrm{bytes}(m(i)) \;\; \forall i \in V_s, \tag{18}$$

i.e. the student vocabulary *is* a 49k-token sub-vocabulary of the teacher's, selected by BPE training on a 545 MB multi-domain corpus restricted to the teacher's token inventory (with a closure pass adding 8,366 tokens so every merge's constituents are present). The map has two consequences measured at build time:

- **Mass coverage.** On held-out teacher-domain text, the teacher assigns on average 98.75% of its next-token probability mass to tokens in the image $m(V_s)$ (5th percentile 95.7%). The teacher's distribution restricted and renormalized to the student vocabulary is therefore a faithful target (§9).
- **Position alignment.** Any student-tokenized sequence maps token-by-token to a valid teacher input $m(x_1), \ldots, m(x_n)$ of the *same length*, so teacher position i scores exactly student position i. Distillation needs no edit-distance or byte-offset alignment machinery at all — the failure class where student and teacher segmentations drift apart is impossible by construction.

Efficiency was not sacrificed. Held-out fertility is 1.0113 tokens/word-piece target, with 98.8% of positions tokenized identically to the teacher; on 1,041 ClimbMix documents the tokenizer achieves 4.637 characters per token — within 3.2% of a reference 128k-vocab tokenizer at 2.6× smaller vocabulary, and at parity with GPT-2's 50k. The 41-check build-time validation suite (round-trips, specials, injectivity, coverage) passed in full.

Table 5: yx49k at a glance.

vocabulary	49,152 (divisible by 128 through 4096)
construction	BPE over teacher-token inventory; injective $m : V_s \to V_t$
teacher	Qwen3.5 (248,077 real tokens; 248,320 padded logit width)
teacher mass in-image	98.75% mean, 95.7% p5
held-out fertility	1.0113; identical-to-teacher positions 98.8%
compression	4.637 chars/token on ClimbMix (−3.2% vs. 128k reference)
specials	33 chat/tool tokens at ids 49119–49151
embedding cost at d=1024	50.3M parameters

Two variants ship: the canonical tokenizer (EOS <|im_end|>) and a pretraining variant binding EOS to <|endoftext|>, keeping the chat terminator out of the base distribution until the SFT stage.

7 The pretraining campaign

7.1 Setup

The model was trained on ClimbMix, a 400B-token filtered web corpus, streamed and tokenized on the fly with a 50k-document shuffle buffer and a 1% held-out validation split. Hardware was a TPU v4-64 slice — 32 chips across 8 hosts — in pure data parallelism: per-device batch 8 at sequence length 2048 gives

$$32 \times 8 \times 2048 = 524{,}288 \text{ tokens per optimizer step}, \tag{19}$$

and the 700,000-step budget totals 3.67×10^{11} *step tokens*. Because the data stream restarts on resume (and the resumed leg re-streams from a fresh shuffle), unique-token exposure is ∼341B — the final 26B

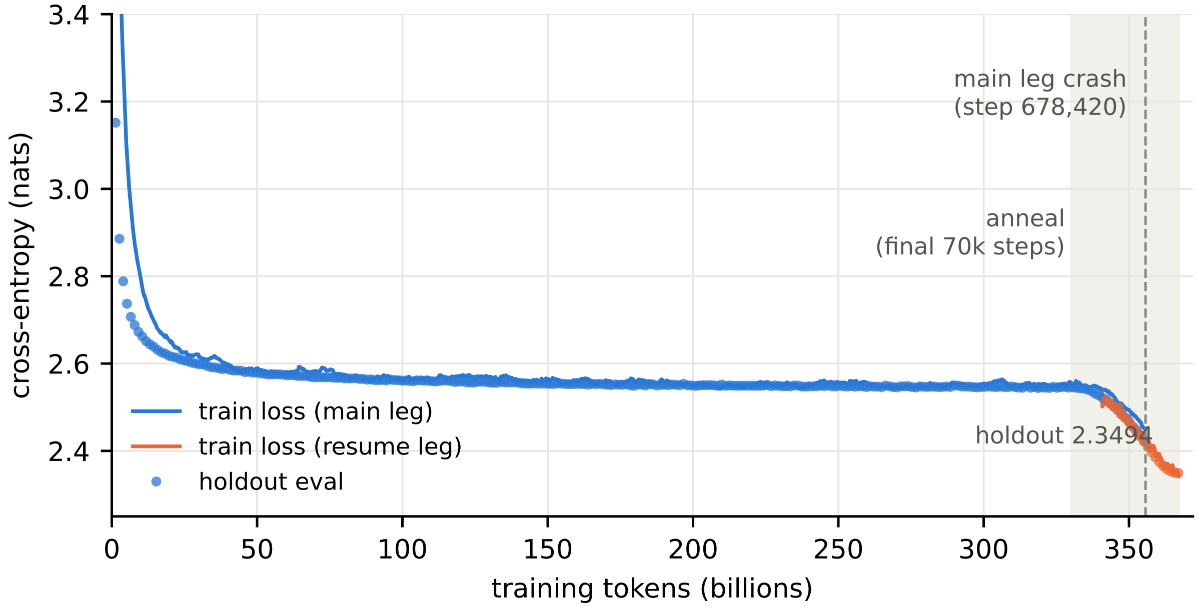

Figure 2: The campaign in one curve: train loss (smoothed) and holdout evaluations for both W&B legs against total training tokens. The shaded band is the terminal anneal; the dashed line marks the main leg's crash at step 678,420. The resume leg (orange) retraces the main leg's trajectory through the overlap region before extending it to 367B tokens and holdout 2.3494.

of step tokens revisit the corpus head. Throughput held at a median of 1.092M tokens/s (480 ms/step, Fig. 4); against the v4 chip's nominal bf16 peak this is a parameter-FLOP utilization of

$$\frac{6\,N\,R}{32 \cdot 275\,\mathrm{TF}} = \frac{6 \cdot 3.077 \times 10^{8} \cdot 1.092 \times 10^{6}}{8.8 \times 10^{15}} \approx 23\%, \tag{20}$$

a lower bound that ignores all sequence-mixing FLOPs. Holdout loss was evaluated every 2,500 steps and a ten-task lm-eval round ran on-device every ∼20k steps, giving the campaign a benchmark trajectory, not just an endpoint (Fig. 5).

7.2 Schedule and anneal

The learning rate followed (17): brief warmup, then a constant 2×10^{-3} for 90% of the run, then cosine to zero over the final 70k steps (Fig. 3). The anneal is where the campaign cashed in: holdout fell from 2.5446 (step 620k, pre-anneal) to 2.3494 — −0.195 nats, roughly the same improvement as the preceding 300k constant-rate steps combined — and every benchmark in the trajectory bent visibly upward through the shaded region of Fig. 5 (HellaSwag 49.3 → 56.9, Lambada 40.4 → 47.4 over the final 80k steps). The benchmark curve was still rising at step 700k; the budget, not convergence, ended the run.

7.3 The crash, and what a clean resume looks like

At step 678,420 — 22k steps from the finish, inside the anneal — the main leg died (W&B state *crashed*; the slice itself survived). Training resumed three days later from the most recent durable checkpoint (≈ step 650k) as a second W&B leg, re-running steps 650k–700k with a restarted data stream.

The resume's fidelity is measurable rather than assumed, three ways. First, the loss curves: through the 28k-step overlap the resumed leg retraces the original to within smoothing width (Fig. 2). Second, the schedule: the resumed learning-rate curve lands exactly on the original cosine (Fig. 3). Third, the benchmark round at step 660k, which both legs executed, agreed within noise (HellaSwag 51.9 vs. 51.3,

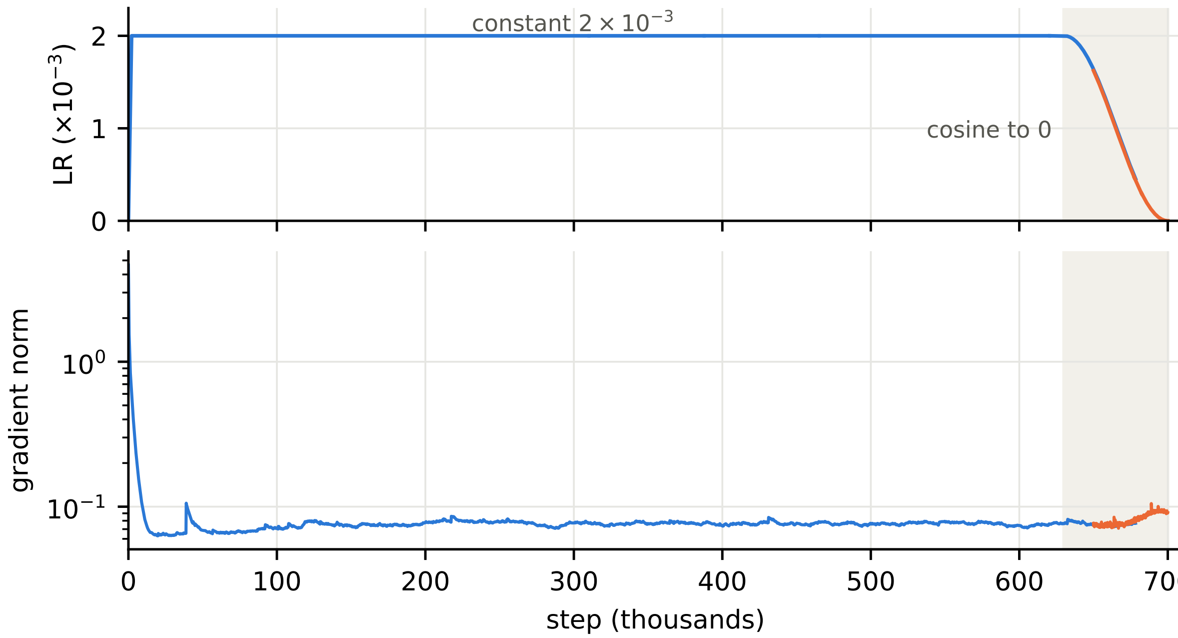

Figure 3: Learning-rate schedule (top) and smoothed gradient norm (bottom, log scale) across both legs. The resume leg's schedule overlays the main leg's exactly through the overlap — the optimizer restored to the same trajectory. The gradient norm is flat at ∼0.07 for the entire constant-rate phase; no clipping events, no spikes, no drift.

ARC-e 63.0 vs. 62.7) despite the different shuffle — the model state, not the data order, determines the trajectory at this point in training. The campaign's conclusion is thereby independent of the crash: the final 700k-step model is the same model the uninterrupted run would have produced, up to data-order noise that the 660k A/B bounds at a few tenths of a point.

8 Base-model evaluation

All numbers below are 0-shot lm-eval results for the step-700k checkpoint unless marked otherwise; reference models are quoted at their published settings.

Table 6: 0-shot suite at step 700k, against same-class references. "prev gen" is this project's predecessor campaign (337M, 40B tokens, 128k-vocab tokenizer).

task	**ours** 308M / 0.37T	SmolLM2 360M / 4T	SmolLM 360M / 0.6T	Qwen2.5 0.5B	Gemma3 PT 270M	prev gen 337M / 40B
HellaSwag	**56.9**	54.5	51.8	51.2	40.9	50.6
ARC avg	**53.8**	53.0	50.1	45.4	43.4	48.9
PIQA	**75.2**	71.7	71.6	69.9	67.7	72.1
OpenBookQA	**38.8**	37.4	37.2	37.4	—	36.2
BoolQ	58.0	—	—	—	61.4	60.1
Lambada	47.4	SciQ 92.0	COPA 72.0	ARC-e/c 67.5 / 40.1		

The headline is the data efficiency: the model beats SmolLM2-360M — the strongest same-size open reference, trained on 4T tokens — on every row where the numbers are comparable, from an 11× smaller token budget. Three further panels complete the picture:

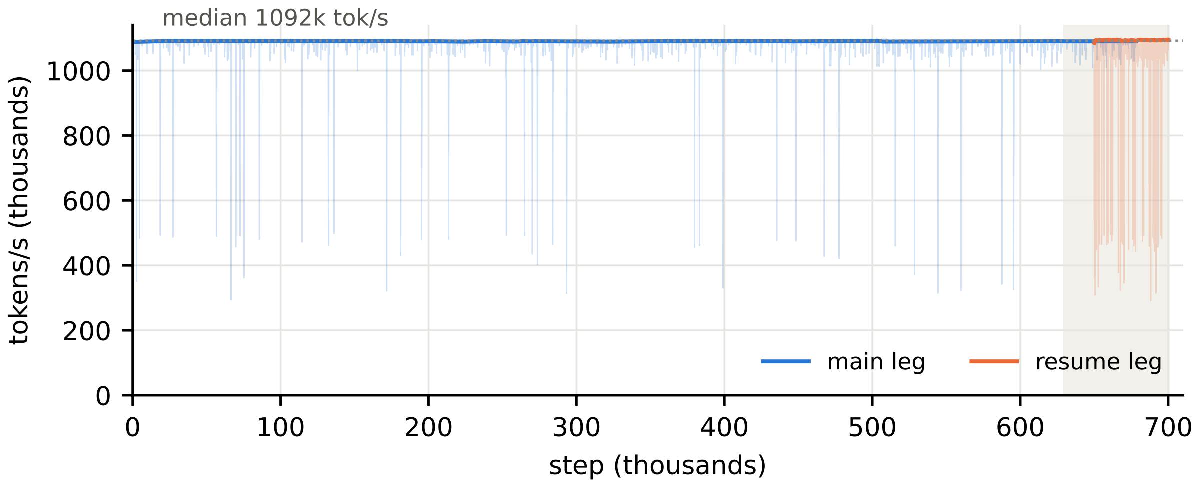

Figure 4: Slice throughput over the campaign (rolling median over raw per-step rates; faint traces show the raw series, whose dips are eval and checkpoint pauses). The line is flat at 1.09M tokens/s for four days of wall time — the data pipeline, running host-side over streamed HF shards, never became the bottleneck.

Table 7: Post-hoc loglikelihood and generation panels (step 700k).

task	ours	best same-class reference	note
WinoGrande 0-shot	**59.4**	SmolLM-360M 52.8	+5 over every reference
WinoGrande 5-shot	**57.9**	Qwen2.5-0.5B 54.1	
MMLU cloze 0-shot	30.0	SmolLM2-360M 35.8	between SmolLM 135M/360M tiers
TriviaQA 5-shot	15.3	SmolLM2-360M 16.9	parity with Gemma3 PT (15.4)
GSM8K 5-shot	1.1	SmolLM2-360M 3.2	base-model band, pre-SFT

WinoGrande is the standout — +5 points over every reference on a task where this size class barely clears chance — and, with HellaSwag and PIQA, suggests the architecture's strength is concentrated in commonsense/coreference resolution. The two knowledge-bound scores (MMLU cloze, TriviaQA) track factual *exposure* rather than capability and sit where an 0.37T-token diet predicts: this is the axis post-training distillation from a knowledge-rich teacher targets next. CommonsenseQA in standard format scores 26.2; small base models cluster at chance there, and published references using cloze formulations are not comparable.

9 Post-training: SFT and offline logit distillation

Post-training proceeded in two stages: supervised fine-tuning to install the chat format, then offline logit distillation from a frozen Qwen3.5-4B teacher to install the teacher's token-level distribution. Every stage decision below was fixed by a measured A/B rather than convention; the ledger records nulls as prominently as wins.

9.1 Data

SFT used the Mephisto instruction corpora together with a large public SFT mix (6.27M rows total, minimum packing length 4096). Two of the Mephisto sets — `Mephisto-IF_172k` (instruction following) and `Mephisto-Knowledge_538k` (knowledge), published under the `Yxanul` organization — were generated on a TPU *v6e* slice with this project's own batched decoding stack; `Mephisto-MathCode_2M` completes the mix with third-party math/code completions. The distillation stage draws prompts from the same pools

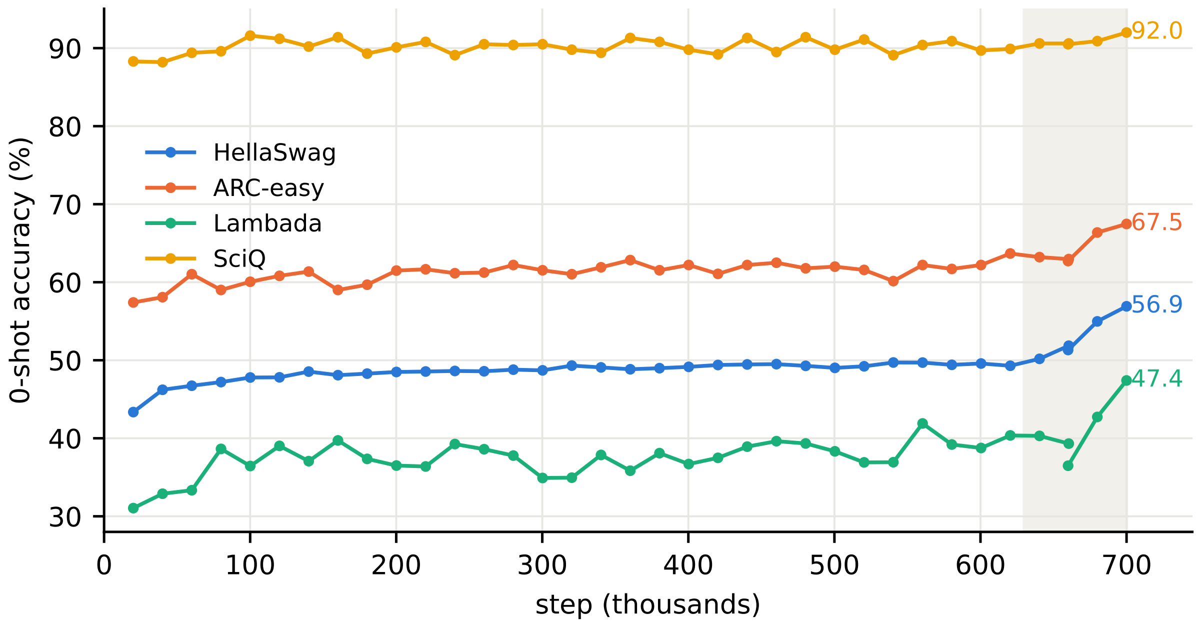

Figure 5: In-training 0-shot benchmark trajectory (both legs), one lm-eval round per ∼20k steps. Long plateaus through the constant-rate phase, then the anneal (shaded) lifts every task simultaneously — the argument, in one figure, for spending the last 10% of any fixed budget on a terminal anneal.

but replaces the completions' one-hot supervision with teacher distributions.

9.2 The distillation objective

Teacher projection. The subset map (18) makes the teacher's distribution directly comparable to the student's. Given teacher logits $z^T \in \mathbb{R}^{|V_t|}$ at a position (truncated to the 248,077 real rows of the padded 248,320-wide head — padded embedding rows emit real logits and must be excluded), the projected teacher log-probabilities on the student support and the residual mass are

$$\ell_i = z^T_{m(i)} - \log \sum_{v \in V_t} e^{z^T_v} \quad (i \in V_s), \qquad \rho = 1 - \sum_{i \in V_s} e^{\ell_i}, \tag{21}$$

computed with a blockwise log-sum-exp (block 32,768, per-block fp32 cast). The residual ρ — the teacher's mass outside the student's image, ∼1.25% on the training mixture per §6 — is *reported and renormalized away*, never trained against. Because m is injective and length-preserving, teacher position i supervises student position i with no alignment pass; the property was verified empirically by checking that shifting the teacher by one position strictly worsens agreement.

Divergence. With renormalized teacher $p_i = e^{\ell_i}/(1-\rho)$ and student $q = \mathrm{softmax}(z^S)$, the general objective is a β-weighted Jensen–Shannon divergence,

$$D_\beta(p \,\|\, q) \;=\; \beta D_{\mathrm{KL}}(p \,\|\, m_\beta) + (1-\beta) D_{\mathrm{KL}}(q \,\|\, m_\beta)\,, \qquad m_\beta = \beta p + (1-\beta) q, \tag{22}$$

with the convention that β weights the *teacher*, so $\beta \to 0$ recovers forward KL $D_{\mathrm{KL}}(p \,\|\, q)$ (mass-covering) and $\beta \to 1$ reverse KL (mode-seeking); the small-β limit $D_\beta \approx \beta D_{\mathrm{KL}}(p \,\|\, q)$ is pinned by a unit test, guarding the orientation. The campaign ran $\beta = 0$ throughout.

Top-K compression and the coverage term. Storing full 49k-wide teacher distributions is wasteful; the pipeline precomputes, once per example, the teacher's top-K ($K \in \{32, 64\}$, ∼90 bytes/position at

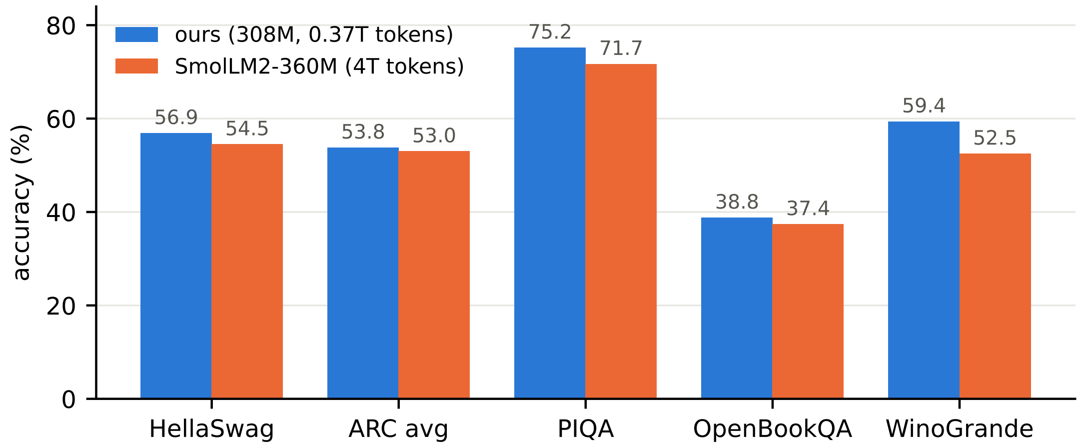

Figure 6: Ours vs. the strongest same-size reference. The model is ahead on every comparable row while having seen $\sim$1/11 the pretraining tokens.

K=32 compressed) via an approximate top-K (recall 0.95; exact top-k lowers to a full sort on TPU and is 87× slower — a missed boundary entry simply stays in the tail bucket). With kept set $\mathcal{K}$, tail $r = 1 - \sum_{i\in\mathcal{K}} e^{\ell_i}$, and renormalized teacher $\tilde{p}$ on $\mathcal{K}$, the β=0 loss uses the student's *raw* probabilities at the kept ids:

$$D \;=\; \sum_{i\in\mathcal{K}} \tilde{p}_i \big(\log \tilde{p}_i - \log q_i\big) \;=\; \underbrace{D_{\mathrm{KL}}(\tilde{p}\,\|\,\tilde{q})}_{\text{truncated forward KL}} \;\; \underbrace{-\log \sum_{i\in\mathcal{K}} q_i}_{\text{coverage}}, \tag{23}$$

where $\tilde{q}$ is the student renormalized on $\mathcal{K}$. The decomposition on the right is the design: an exact truncated forward KL plus a penalty on student mass leaking outside the teacher's top-K — non-negative, and zero exactly when the student matches the renormalized teacher on $\mathcal{K}$ and carries no outside mass. At $K = |V_s|$ the compression is exact and (23) reproduces the full-support loss (also pinned by test). The composite objective $\mathcal{L} = w_d D + w_{ce}$ CE shares one masked-position denominator between both terms; the campaign ran pure distillation ($w_d = 1$, $w_{ce} = 0$) with CE logged as a monitor.

9.3 Results ladder and the epoch null

Table 8: The post-training lineage. IFEval is the mean of the four strict/loose prompt/instruction accuracies; panel is a 42-prompt scored transcript battery. All decoding at each checkpoint's swept settings.

checkpoint	recipe	IFEval	panel /42
gen-2	base + SFT 690k rows	40.4	10
GOLD-10k	gen-2 + distill 10k IF	42.1	—
GOLD-50k	gen-2 + distill 50k IF	48.4	12
GOLD-mix300k	gen-2 + distill 100k×{IF, Know, MathCode}	**51.0**	16
SFT-6M	base + SFT 6.27M rows	35.3	15
GOLD-over-SFT	SFT-6M + distill 1/3 mix, K=32	41.3	**22**
(2-epoch control)	same store, 2 epochs	40.3	22

Three findings organize the ladder. *Distillation dominates SFT per row*: 300k distilled examples beat 6.27M one-hot rows by +15.7 IFEval — soft targets carry more supervision per example than hard labels by more than an order of magnitude at this scale. *Mixed diets beat matched diets*: distilling knowledge and math/code rows alongside instruction data improved *instruction following* beyond what pure-IF distillation achieved. *One epoch per store*: a controlled second epoch over the same precomputed targets moved the

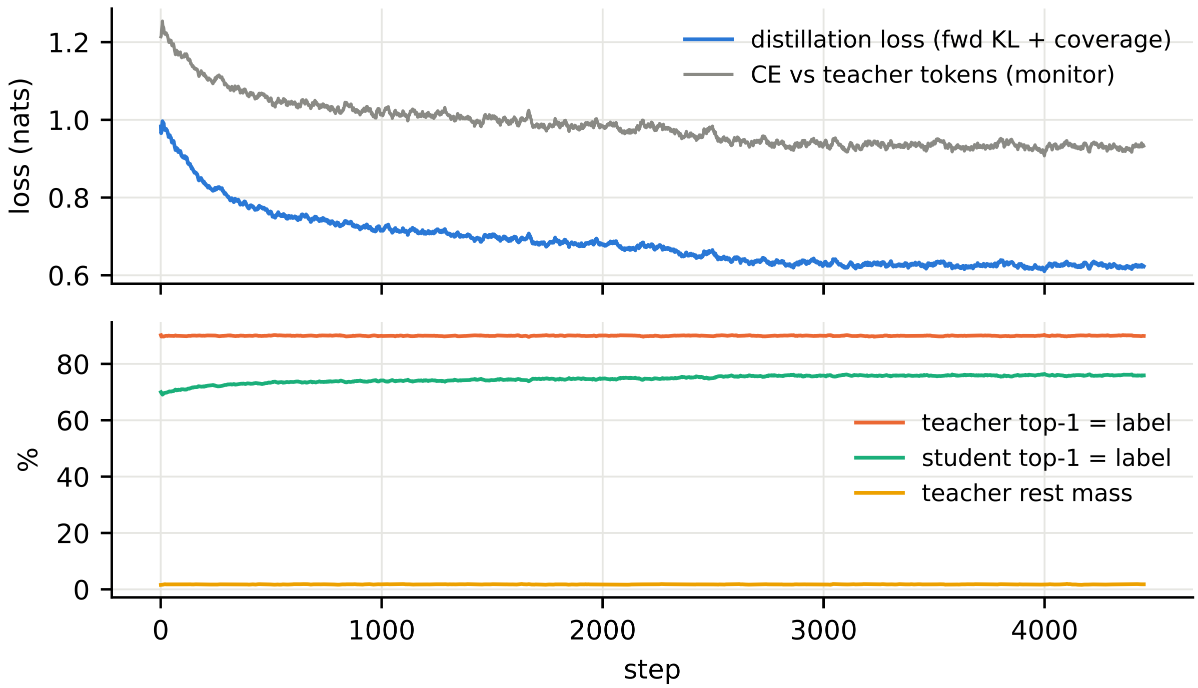

Figure 7: The 300k-example distillation run (top: losses; bottom: top-1 agreement and residual mass). The distillation loss falls 37% while the CE monitor barely moves — the student is learning the teacher's *distribution*, not memorizing tokens. Teacher top-1 agreement with the data's labels sits at 90% (an upper reference), student agreement climbs toward 76%, and the teacher's rest mass holds near 1.5%.

training loss (0.504 → 0.480) and nothing downstream (IFEval 41.3 → 40.3, panel 22 → 22) — the marginal unit of compute belongs to fresh rows, not repetition.

10 External calibration

Published numbers do not transfer across evaluation harnesses, so the final calibration measured every model under this project's own conventions: identical prompts, identical scoring code, IFEval with the same chat-template and generation settings, and each model decoded at its documented or swept-best settings. The references are the three strongest openly available instruction-tuned models near 300M: Gemma 3 270M IT, SmolLM2-360M-Instruct, and SmolLM-360M-Instruct.

The matrix separates two axes cleanly. On *capability* (the loglikelihood rows) our model leads 3/4 against every reference, from 11–16× less pretraining data; only SmolLM2's 4T tokens buy it one row (HellaSwag). On *behavior* (the transcript panel) the deeply post-trained references lead: SmolLM2's mature SFT+preference pipeline yields drilled math/code answer rituals (8/8 and 8/8 on those panel sections vs. our 4/8, 4/8) and disciplined termination, while our transcripts' signature failure is producing the *correct* answer and then destroying it in continued generation — computing 144 and talking itself to 1440, repeating a birthday greeting thirty-six times. Notably, our IFEval still leads all references (mix300k's 51.0 is +10.9 over the best measured reference), and the same-family SmolLM v1→v2 jump (panel 19 → 34, IFEval 16.1 → 40.1 at identical parameter count) is an existence proof that everything separating the columns is data and post-training recipe — not architecture, and not size.

The failure signature also names the cure. Right-answer-then-drift is textbook exposure bias: teacher-forced training (SFT and offline distillation alike) never shows the model its own repetition spirals, so nothing ever penalizes them. On-policy distillation — sampling from the student and letting the teacher score the student's own continuations — attacks exactly this, and is the designated next stage.

Table 9: Everything measured, one harness. Panel: 42 scored prompts (knowledge/math/code/instruction-format). Chat: 12 freeform prompts with lenient checkers. Loglik: measured 0-shot (hellaswag / arc-c / piqa / winogrande). IFEval: mean of the four accuracies.

	SmolLM-IT 360M / 0.6T	Gemma3 IT 270M / 6T	**ours** 308M / 0.37T	SmolLM2-IT 360M / 4T
panel /42	19	31	22	**34**
chat /12	10	8	9	**11**
IFEval (GOLD-over-SFT / mix300k)	16.1	31.4	41.3 / **51.0**	40.1
loglik: hellaswag	52.7	37.7†	53.5	**56.8**
loglik: arc-challenge	33.0	28.2†	**37.5**	34.0
loglik: piqa	70.6	66.2†	**73.4**	71.3
loglik: winogrande	53.6	52.3†	**58.0**	57.6

†Gemma's published IT numbers (0-shot, matching tasks); its measured IFEval of 31.4 sits far below its published 51.2 — the concrete case for measuring everything under one harness.

11 Discussion and future work

What the evidence supports. At the 300M scale, this program found (i) architecture and data quality dominate raw token count — a 0.37T-token hybrid beats a same-size 4T-token reference on most capability measures; (ii) the terminal anneal is the highest-leverage 10% of a fixed pretraining budget; (iii) soft-target distillation carries over an order of magnitude more supervision per example than one-hot SFT; and (iv) the remaining gap to the best same-size instruction models is decode-time discipline — a post-training property, purchasable at this size, as the SmolLM v1→v2 delta demonstrates.

Limitations. The base model's knowledge-bound axis (MMLU, TriviaQA) reflects its token budget and remains below the 4T-token reference. Strict-format instruction following is hard for every model measured at this size; the best reference reaches 5/8 on that panel section where we reach 3/8. The repetition spiral survives offline distillation; whether it survives on-policy training is the next experiment's question, and the honest possibility remains that some of it is a capacity effect. All pretraining ran at sequence length 2048; the architecture's long-context economics (Fig. 1) are demonstrated for throughput but not yet exercised by training.

Next. In order of expected value: an on-policy distillation stage (student-sampled rollouts scored by the teacher — directly targeted at the observed failure class); the full offline campaign over the remaining prompt pools at K=32 with termination-weighted data; and a width/depth scale-up for which the measured groundwork (an 884M configuration's loss-head and sharding sweeps) already exists in the repository. The comparison battery of §10 is fully scripted and will be re-run unchanged after each stage; the bar it sets is precise — match SmolLM2-360M-Instruct's 34/42 transcript discipline while keeping the capability and IFEval leads — and clearing it would make this the strongest model of its size class outright.

Acknowledgments. Compute was provided by the TPU Research Cloud. The pretraining corpus is ClimbMix; evaluation uses the EleutherAI lm-evaluation-harness; the distillation teacher and the tokenizer's parent vocabulary are Qwen3.5.

A Kernel experiment ledger (digest)

The kernel program ran as 42 numbered experiments over nine days on a TPU v6e-8, then a v4-64. The decision path, compressed:

EXP-001..008	Baselines: 272M dense transformer, INT8 rejected, modern GQA recipe, orthogonalized-momentum optimizer adopted.
EXP-009..015	KDA implemented (reference recurrence, chunkwise form); 3:1 hybrid matched for throughput; external kernel codebases audited — only the state-scan skeleton proved reusable.
EXP-016..023	The backward problem: blocked Pallas solve, analytical XLA VJP (31.9 ms), then the fused Pallas pair (18.3 ms core); grid collapse and selective bf16 (§4) reach 616k tok/s full-model.
EXP-024..028	Batch/remat/sequence sweeps; the 16k crossover of Fig. 1.
EXP-029..037	Real-data gates: 10B-token smoke, precision-on-real-text audit (kills repeated squaring), 1B-token gradient gate, stable substitution fallback, divide-and-conquer inverse adopted (+30%), conv folding, fused linear CE (v6e only).
EXP-038..042	v4-64 era: optimizer matricization decompositions, WY-conditioning statistics on real text, 3-pass emulated solve qualified, output-gate A/B (neutral), learning-rate sweep (the 10× under-tuning finding).

B Configuration dump

Table 10: Campaign configuration (values as launched).

model	`kda_hybrid_yx49k_l20`	optimizer	MuonClip
steps	700,000	peak LR	2×10^{-3}
anneal	cosine, final 70k steps	warmup	brief linear
seq len	2048	per-device batch	8
tokens/step	524,288	grad clip	1.0 (global ℓ_2)
weight decay	0.1 (both arms)	NS steps	5
Muon β	0.95	AdamW $\beta_{1,2}$	0.9, 0.95
QK-clip τ	100	consistent-RMS	0.2
KDA precision	`guarded_fp32`	solve	inverse, 3-pass (v4)
remat	minimal-with-context	loss	standard CE, no z-loss
data	ClimbMix, streamed	shuffle buffer	50,000 docs
holdout	1%, eval every 2,500	lm-eval	10 tasks / ∼20k
checkpoints	every ∼19k steps, keep 6	mesh	32-way data parallel

C Evaluation conventions

Loglikelihood tasks use lm-eval defaults with each task's conventional primary metric (acc_norm where defined). IFEval runs generatively with the chat template, 512 max new tokens, and each model's documented or swept decode settings; all four accuracies (prompt/instruction × strict/loose) are reported and the mean is used as the scalar. The 42-prompt transcript panel scores programmatic checkers (code is executed against assertions; formats are matched exactly); each model was measured at greedy *and* its shipped sampling settings, quoting the better. Externally-published numbers appear only where a task was not re-measured, and are marked as such wherever they occur.